# Wei Pan Ph.D.

R&D Director · Industrial Machine Vision
willpan.xyz · vpan@foxmail.com · +86 135 3812 8621 · Tokyo / Dongguan

**100+** Published patents **20+** Algorithm team **12** Top-tier papers **5** Industries in production

## Summary

Eight years in industrial machine vision and 3D imaging; SUTD Ph.D. Currently R&D Director at OPT Tokyo, leading a 20+ algorithm team. Focused on high-precision 3D imaging (structured light, phase deflectometry, photometric stereo), point cloud / mesh processing, industrial defect inspection, and robotic vision — deployed in Li-battery, 3C, automotive, medical, and logistics production lines.

## Experience

2013 – present · academia and industry, in parallel

**2013 Committed to 3D geometry**
Entered ultra-scale rapid prototyping and assembly planning at SUTD, locking "geometry + algorithm" as the long arc.

**2019 From academia to industry**
Joined OPT during postdoc — owned structured-light reconstruction and SciVision 3D SDK, turning algorithms into shipping code.

**2021 IC to leadership**
Became R&D Director at OPT Tokyo, shifting from individual contributor to team leader, driving funded national R&D programs and cross-industry production rollouts.

### OPT Tokyo R&D Center · industrial vision algorithm stack — R&D DIRECTOR

ROLE Director at OPT Tokyo since 2021. Built a 20+ algorithm team from zero, owning the full industrial vision algorithm stack and coordinating between Japan field clients and the China HQ R&D org.

ACTIONS Designed three product lines (3D imaging / defect inspection / robotic guidance); shipped fused structured-light, phase-deflectometry and photometric-stereo pipelines; led joint research between Tokyo applied team and Dongguan HQ.

IMPACT Powers the full SciVision 3D camera algorithm stack; 100+ published patent applications, 25+ granted, 2 PCT; deployed across five industries: Li-battery, 3C, automotive, medical, logistics.

### Dongguan Major Sci-Tech Program · high-end manufacturing inspection — PRINCIPAL INVESTIGATOR

ROLE PI of the 2023–2025 Dongguan Major Sci-Tech Program "Intelligent Inspection for High-end Manufacturing", a joint university-industry effort centered on structured-light and AI-based defect algorithms.

ACTIONS Coordinated algorithm / hardware / application teams; delivered fused phase-deflectometry plus photometric-stereo inspection, depth-map jitter correction, and semantic point-cloud registration.

IMPACT Within the cycle: 30+ patents filed, 1 PCT, and 4 SCI papers including EAAI 2026; algorithms in production for battery sealing-nail defect, glass coating evaluation, and automotive weld-quality inspection.

### 3D Imaging Products & SciVision 3D SDK · core company product line — MODULE OWNER

ROLE Owner of the company's 3D imaging products and the SciVision 3D SDK core modules since 2019 — multi-modal imaging, depth / point-cloud / mesh SDK, and real-time rendering, architecture through shipping.

ACTIONS Led high-precision reconstruction across structured light, line laser, and phase deflectometry; shipped depth / point-cloud / mesh filtering, CAD registration, and height/volume measurement SDK modules; authored original algorithms — HLO, Chebyshev flatness.

IMPACT Powers the company's full 3D camera line (structured light, line laser, etc.); HLO (half-kernel Laplacian) published in CAD; granted patents cover point-cloud normals (CN112802077B), progressive rendering (CN113269860B / WO2022257594), and mesh filtering (CN110120069B / CN113178013B).

### Robotic 3D Vision Grasping · Dongguan Key R&D Project — KEY MEMBER

ROLE Key member of the 2020–2022 Dongguan Key R&D Project "Robotic 3D Vision Smart-Grasping System", responsible for hand-eye calibration and 3D object recognition algorithms.

ACTIONS Proposed a 3D-object-recognition-based hand-eye calibration method; built an automated 3D scanning rig for UR arms, closing the loop of "scan · recognize · grasp".

IMPACT Demonstrated in logistics sorting; granted patent CN109702738B (3D recognition-based hand-eye calibration) and related patent CN109141281A (3D scanning device); folded into the company's robotic-guidance product line.

## Education

**SUTD, Singapore** 2013–2018
Ph.D. in Engineering Product Development · advisor Prof. Lujie Chen · Presidential Graduate Fellowship

**Zhejiang University** 2009–2013
B.Eng. in Optical Engineering · Chu Kochen Honors College · GPA 3.60 / 4.0

## Selected Publications

2020 - 2026 · ★ marks first / corresponding author · all SCI Q1 or top venues

| | | |
|---|---|---|
| EAAI | ★ Improved graph attention network for industrial point clouds: battery sealing-nail defect segmentation. | 2026 |
| Patterns · Cell | ★ Deep learning-enabled 3D multimodal fusion of CBCT and intraoral mesh scans. | 2023 |
| Opt. Lasers Eng. | Line-scan calibration from elliptical distortion of circles. | 2026 |
| Optics Express | Transformer-enhanced structured light for moving objects. | 2025 |
| Optics Express | Reconstruction of transparent objects using PSP based on diffusion models. | 2024 |
| Meas. Sci. Tech. | ★ Gap measurement via projection lines and convex analysis of 3D point cloud. | 2024 |
| ECCV | SemReg: Semantics-constrained point cloud registration. | 2024 |
| EAAI | ★ Physically inspired geometry-to-photometry rendering for real-time weld defect detection. | 2026 |
| CVPR · Highlight | SuP: sub-cloud driven point cloud registration. | 2026 |
| Opt. Laser Technol. | Depth-adaptive calibration of line-laser 3D sensors. | 2026 |
| Adv. Photonics | Generative deep-learning-embedded asynchronous structured light for 3D imaging. | 2024 |
| CAD | ★ HLO: Half-kernel Laplacian Operator for Surface Smoothing. | 2020 |
| IEEE TVCG | DeSC: Learning Deep Semantic Descriptor for NeRF Registration (in press). | 2025 |

## Selected Patents

granted / published · 100+ applications · 25+ granted · ★ marks first inventor

| | | |
|---|---|---|
| CN119006550B | ★ Neural Radiance Field (NeRF) registration method & system | 2025 |
| CN114324168B | Surface defect detection method & system | 2024 |
| CN113269860B<br>WO2022257594 | ★ High-precision real-time progressive 3D rendering · PCT | 2021 |
| CN112802077B | ★ High-precision point cloud normal estimation | 2021 |
| CN113178013B | ★ Triangle mesh filtering method & device | 2021 |
| CN110120069B | ★ Laplacian-based triangle mesh filtering | 2019 |
| CN120598883A | ★ Fused phase-deflectometry + photometric-stereo defect inspection | 2025 |
| CN120388003A | ★ Multi-cost speckle structured-light stereo matching | 2025 |
| CN120339358A | ★ Fused phase-deflectometry + photometric-stereo surface reconstruction | 2025 |
| CN119919533B | ★ Downsampled single-frequency phase unwrapping | 2025 |
| CN118397020A | ★ Fast image region segmentation and contour extraction | 2024 |
| CN109702738B | 3D recognition-based robotic hand-eye calibration | 2019 |

## Service & Awards

### Academic Service

| | |
|---|---|
| JOURNAL REVIEWER | EAAI · ESWA · AUTCON · CVIU · KBS · TVCG · IJCV · CVMJ · Scientific Reports |
| CONFERENCE REVIEWER | NeurIPS 2026 · CVPR 2025/26 · ECCV 2025/26 · ACM MM 2024/25/26 · AAAI 2026 |
| PROGRAM COMMITTEE | ICICV 2025 · ICIGP 2025 / 2026 · ACM MM 2024 · MCACCV 2022 |
| EDITORIAL BOARD | The Computational Vision and Imaging |

### Selected Awards

| | | |
|---|---|---|
| 1ST PRIZE | Guangdong Machinery Industry Sci-Tech Award | 2023 |
| 1ST PRIZE | Guangdong Mech. Eng. Society Sci-Tech Award | 2023 |
| 2ND PRIZE | Henan Province Sci-Tech Progress Award | 2023 |
| PHD FELLOWSHIP | SUTD Presidential Graduate Fellowship (5 yr) | 2013–18 |

## Core Skills

| | |
|---|---|
| 3D IMAGING | structured light · line laser · phase deflectometry · photometric stereo · speckle stereo; high-speed reconstruction and multi-modal fusion |
| POINT CLOUD & MESH | filtering, normals, registration, segmentation, CAD matching, feature extraction; original HLO · half-window Laplacian · Chebyshev flatness |
| DEFECT INSPECTION | deep learning + classical fusion; welding · sealing · surface · QR / DM code · contour; battery sealing-nail and automotive weld in production |
| ROBOTIC VISION | 3D-guided grasping, hand-eye calibration, object recognition; UR arm productionization, end-to-end "scan · recognize · grasp" |
| ENGINEERING & LEADERSHIP | C++ / Python / MATLAB; leads 20+ team; PI on government-funded programs; cross-site coordination across Tokyo and Dongguan |

*"Less, but better."* — Dieter Rams

*"Perfection is achieved not when there is nothing more to add, but when there is nothing left to take away."* — Saint-Exupéry

*"Constraints are catalysts."*